

Precision–Recall Trade-offs in Radar-Only, Camera-Gated, and Emergency-Fallback AEB: A Repeatable CARLA Study

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Abstract—A hard camera permission gate can suppress radar-triggered false braking, but it can also veto genuine hazards outside the camera detector’s label space. This paper quantifies that trade-off and evaluates a conservative radar emergency fallback in a sensor-to-brake automatic emergency braking (AEB) pipeline in CARLA 0.9.11. Radar detections are path-filtered, clustered, tracked, and used for time-to-collision and stopping-margin decisions; YOLO26n supplies *car* confirmation. The fallback permits an unconfirmed radar brake only for a persistent, well-supported central track with simultaneously critical TTC and stopping margin, then latches until radar brake release. A frozen GPU campaign contains 2,461 runs in 639 process-isolated CARLA sessions. On a constructed 525-run core benchmark excluding labelled synthetic faults, radar-only, hard-gate, and fallback policies respectively obtain precision/recall of 0.913/0.988, 1.000/0.965, and 1.000/0.988; fallback reduces hard-gate collisions from 25 to 15. This apparent combination does not generalize as dominance. On a 70-run frozen hold-out, fallback precision falls to 0.600 because convincing 6–10-point synthetic ghosts trigger 20 false brakes; all policies collide in 15 runs involving three unseen physical props. Ablation, perturbation, camera disabling, and CUDA timing identify both useful constraints and unresolved failure modes. The results support camera gating as false-positive suppression and emergency fallback as a conditional trade-off, not as a general obstacle solution or safety certification.

Index Terms—automatic emergency braking, camera–radar fusion, CARLA, false braking, precision–recall, fault injection, hold-out evaluation

I. Introduction

Automatic emergency braking is a closed-loop perception–decision–actuation problem. Radar directly supplies range and radial velocity, but a selected return need not be a genuine obstacle in the ego path. Camera semantics can reject implausible radar targets, but a missed or out-of-label object is not proof that the road is clear. AEB reviews consequently treat perception, decision logic, and control as an interdependent chain [1]. Radar–vision vehicle recognition and AEB-oriented association are established ideas [2], [3]; this study does not claim a new fusion primitive.

Simulation enables repeatable near-collision experiments [4], [5], yet two methodological traps remain. First, simulator actor identity or perfect kinematics can turn an end-to-end AEB test into an ideal-state controller test. Second, repeatedly confirming a rule on scenarios used to design it can hide

its failure boundary. We therefore use CARLA radar points and RGB frames for online decisions, reserve actor truth for scoring, freeze the policy before hold-out, and retain all unsuccessful cases.

Earlier work from this project found a clear asymmetry: hard camera gating removed 40/40 false brakes on non-hazard edge props, but missed central box and barrel hazards in 10/10 runs. The present work asks whether a radar emergency fallback can recover the latter without discarding the former. Its contributions are: (1) a fixed three-policy comparison—radar-only, hard camera gate, and camera gate with emergency fallback; (2) component ablation and one-factor sensitivity; (3) perturbation, detector-disabled, and frozen hold-out suites; and (4) a resumable GPU protocol that logs active inference providers and starts a fresh CARLA server for each named scenario. The central result is deliberately negative as well as positive: fallback combines the two core metrics, but strong radar ghosts and unseen physical props prevent a dominance claim.

II. Related Work and Positioning

Radar–camera AEB has been studied at recognition, association, decision, and actuation levels. Kim and Song combine radar and vision for vehicle recognition [2]; Hsu *et al.* address radar/camera noise filtering for AEB target selection [3]. Bae *et al.* estimate the closest in-path vehicle through low-channel LiDAR projection and camera tracking on a real vehicle [6]. Zhang *et al.* combine curved-road target recognition, TTC, safety distance, and graded braking in CarSim/Simulink [7]. These studies establish that semantic verification, path relevance, and staged intervention are not individually new.

CARLA has supported full autonomous-pipeline validation under protocol-derived scenarios [8]. Its value here is not merely visualization: a fixed-tick sensor-to-actuator loop permits collision-inducing tests, complete tick logs, and repeatable policy substitution. Its limitation is equally important. Simulator radar abstractions, graphics, map geometry, and actor catalogues define a synthetic domain, while a simulator can expose privileged truth unavailable to a deployed AEB. This work therefore separates online sensor information from offline outcome truth and reports both system success and detector/radar failure modes.

Table I states the resulting position. Relative to prior work, the contribution is not a higher-capacity detector or probabilistic tracker. It is a controlled comparison of three late-decision policies plus a frozen adverse hold-out. In particular, the hold-out asks whether a rule created to recover weakly classified objects also accepts a radar fault that has the same rule-level attributes. This falsification-oriented evaluation distinguishes the study from a demonstration on nominal examples.

III. System and Policies

A. Sensor-to-brake information boundary

The evaluated system uses a Tesla Model 3 on Town04, an RGB camera and a 100-m forward radar at a 0.05-s sensor tick. CARLA radar is a point-detection model, not a raw FMCW signal chain; virtual radar fidelity depends on model level [9]. Exact sensor transforms calibrate radar-to-image projection, and map height supports road-ground rejection. Online braking does not query actor ID, actor velocity, class ground truth, or ground-truth boxes. Those quantities are used only to label and score runs.

Radar points are rejected by range, height, road-ground, and a predicted constant-curvature path. Remaining points are clustered by spatial and velocity proximity and associated across frames. A selectable track must be confirmed, non-stale, and collision-relevant. For selected distance d and radar relative velocity $v_r < 0$,

$$v_c = -v_r, \quad \text{TTC} = d/v_c. \quad (1)$$

With ego speed v_e and $v_t = \max(0, v_e + v_r)$, the stopping model is

$$d_{\text{req}} = \max\left(d_0, v_e t_r + \frac{v_e^2}{2a_e} - \frac{v_t^2}{2a_t} + d_0\right), \quad m = d - d_{\text{req}}, \quad (2)$$

where $t_r = 0.20$ s, $a_e = 8$ m/s², $a_t = 6$ m/s², and $d_0 = 1$ m are simulation risk parameters. A staged PID maps risk to continuous CARLA brake with soft, medium, hard, and emergency bounds. The controller and its binary ablation have been evaluated separately; staged PID avoids five additional collisions in a 330-run system-limit sweep.

YOLO26n was fine-tuned for one *car* class using CARLA images. Its held-out in-domain precision, recall, mAP₅₀, and mAP_{50:95} are 0.9919, 0.9642, 0.9940, and 0.9495. These detector scores establish narrow in-domain suitability, not AEB safety. For target camera coordinates (X_c, Y_c, Z_c) ,

$$u = f_x X_c / Z_c + c_x, \quad v = f_y Y_c / Z_c + c_y, \quad Z_c > 0. \quad (3)$$

The camera confirms a radar target when (u, v) lies inside a YOLO car box; the confirmation is held for 0.35 s of simulation time.

The online order is fixed: update the ego path, filter radar points, update tracks, select one target, compute radar risk, infer the latest camera frame, apply one of the three permission policies, actuate, and log the next simulator state. Camera boxes cannot create kinematic risk without a radar target. Radar remains the source of distance and relative velocity.

This ordering matters: the policies differ only at the late permission gate, while radar processing, risk equations, staged PID, scenario control, and scoring remain paired.

B. Three brake policies

Radar-only passes the radar AEB decision directly. The hard gate permits **BRAKE** only with current or held camera confirmation. The proposed fallback policy is

$$B_f = B_r \wedge (C \vee H \vee E_r), \quad (4)$$

where B_r is radar brake, C is current confirmation, H is confirmation hold, and E_r is emergency radar evidence. The latter requires a confirmed, non-stale track with age and hit streak ≥ 3 , at least six cluster points, confidence ≥ 0.70 , predicted-path offset ≤ 0.65 m, $\text{TTC} \leq 1.10$ s, and $m \leq -2.0$ m. TTC and margin must both be critical. Once accepted, fallback is latched until radar leaves **BRAKE**. No actor type or scenario label is used. This rule limits radar override; it cannot prove that a well-supported radar track is physically genuine.

IV. Frozen Experimental Protocol

The simulator runs synchronously at 0.05 s with physics control. The v1 policy and evaluation protocol were tagged before hold-out. Hard gate and fallback use the same ONNX model, 640-pixel input, 0.15-s simulation-time cadence, and a required `CUDAExecutionProvider`; CPU fallback or inference error is a technical hard stop. A master runner checkpoints every run, executes all repetitions of one named scenario, terminates CARLA, then starts a fresh Town04 server. Algorithmic failures do not stop the campaign.

The 2,461-run campaign comprises: 22 smoke runs; 324 development ablation/sensitivity runs; 1,665 core runs (555 per policy); 180 local perturbation runs; 60 detector-disabled runs; and 210 hold-out runs. Core contains a repeated 66-case CCRm/CCRb/cut-in system-limit sweep, a mixed vehicle/negative regression suite, 40 edge-prop runs, 10 central box/barrel hazards, and 30 explicitly labelled four-point synthetic radar faults. The faults are injections, not native CARLA ghost statistics.

Twenty perturbation conditions change ego speed by ± 2 km/h, gap by ± 1 m, cut-in start by ± 0.1 s, and box lateral position by ± 0.15 m. Detector-disabled tests emulate no camera detections. The frozen hold-out contains new shopping-cart/bench/traffic-warning props, four vehicle appearances, two edge props, and five persistent synthetic ghosts with 6–10 points. Hold-out results are not used to retune v1.

A positive label means braking is expected; a positive prediction means brake was activated. Collision and full scenario PASS are reported separately because a true-positive brake can still occur too late. Precision/recall are composition-dependent: suite class ratios are deliberately designed, not road prevalence. Wilson intervals describe run-level proportions, while all-PASS/all-FAIL/mixed outcomes are also inspected by named scenario. Repeats measure consistency and are not treated as independent traffic encounters.

TABLE I
Positioning against representative in-path perception and AEB studies.

Study	Perception / association	Decision / control	Evaluation emphasis
Kim and Song [2]	Radar-vision vehicle recognition	AEB-oriented target recognition	Sensor-fusion recognition
Hsu <i>et al.</i> [3]	Radar/camera noise filtering	AEB target selection	Noise-filter design
Bae <i>et al.</i> [6]	LiDAR projection and camera tracking	Closest in-path vehicle for AEB	Real-vehicle scenarios
Zhang <i>et al.</i> [7]	Curved-road hazardous-target selection	TTC, safety distance, graded braking	CarSim/Simulink
This work	CARLA point-to-track radar and YOLO permission gate	Radar-only, hard gate, emergency fallback; staged PID	2,461 runs; ablation, degradation, perturbation, frozen hold-out

TABLE II
Frozen emergency-fallback configuration.

Condition	Value
Track age / hit streak	$\geq 3/3$ frames
Cluster points / confidence	$\geq 6/0.70$
Absolute path offset	≤ 0.65 m
TTC / stopping margin	≤ 1.10 s / ≤ -2.0 m
Risk relation	TTC AND margin
Release	Latch until radar non-BRAKE

TABLE III
Core benchmark excluding synthetic fault injection (525 runs/policy).

Policy	TP	FP	TN	FN	Prec.	Rec.	Coll.
Radar-only	420	40	60	5	.913	.988	15
Hard gate	410	0	100	15	1.000	.965	25
Fallback	420	0	100	5	1.000	.988	15

V. Results

A. Core trade-off

Table III excludes synthetic fault injection from the primary core confusion matrix. Radar-only has higher recall than hard gate but 40 edge-prop false brakes. Hard gate has no false positives but misses ten central non-vehicle runs in addition to five late cut-out runs. Fallback recovers the ten central runs while retaining zero core false positives. Its 15 collisions equal radar-only and are ten fewer than hard gate. The 95% Wilson intervals are 0.884–0.936 for radar precision, 0.973–0.995 for radar/fallback recall, 0.943–0.979 for hard-gate recall, and 0.991–1.000 for both zero-FP precision estimates. Adding the 30 weak synthetic faults changes radar precision to 0.857; both camera policies block all 30.

Across all 555 core runs, fallback improves 70 radar-only system outcomes and regresses none; it improves ten hard-gate outcomes and regresses none. This is a within-core statement. The frozen hold-out reverses the hard-gate comparison. Fig. 1 isolates three deliberately constructed stresses. Radar passes the central props but brakes on every edge prop and weak synthetic fault; hard gate has the opposite central-prop outcome; fallback passes all three core stress families.

B. Ablation, sensitivity, and degradation

Fig. 3 visualizes system-level PASS rates for the component removals. It should not be read as equal statistical evidence for every rule: the focused scenarios are designed to expose

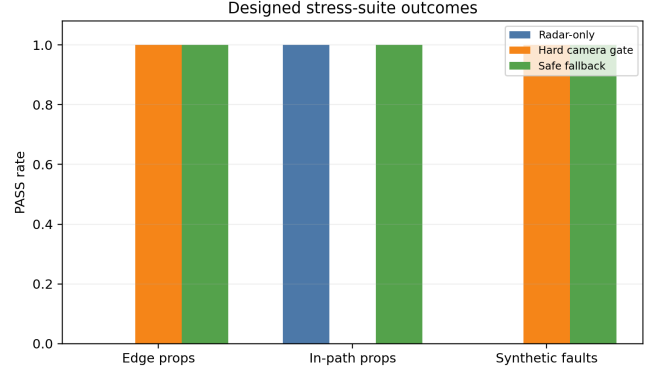


Fig. 1. Core stress-suite PASS rates. These suites are balanced mechanisms, not a prevalence-weighted sample.

path, point-count, and latch mechanisms, whereas all tracks are persistent and strongly critical.

The focused ablation uses 12 development scenarios repeated twice. Full fallback passes 24/24. Hard gate and no-latch each fail four central box/barrel runs; removing the central-path constraint creates six edge-prop false brakes; removing the point threshold creates four weak synthetic false brakes. Removing stability/confidence or changing TTC-and-margin to TTC-or-margin does not change this focused subset, so independent necessity is not claimed for those rules.

One-factor sensitivity uses six representative scenarios repeated twice. The nominal policy passes 12/12. Reducing minimum points from six to four creates two synthetic false brakes. Increasing stability from three to four frames or tightening TTC from 1.10 to 0.90 s misses the central box twice. Tested path, margin, confidence, and permissive-TTC alternatives do not change this subset. These results expose a local precision–recall surface rather than optimize it.

In perturbation tests, hard gate misses all five box conditions; fallback recovers three, but collisions remain at 19- and 21-m gaps despite passing the 20-m nominal condition. Radar-only also collides at 19 m. This non-monotonic outcome reflects return geometry, track confirmation time, and stopping margin. With the detector disabled, fallback activates on all 18 positives but still collides in six stationary/braking-lead runs because emergency evidence arrives too late. Brake recall therefore does not imply safe stopping.

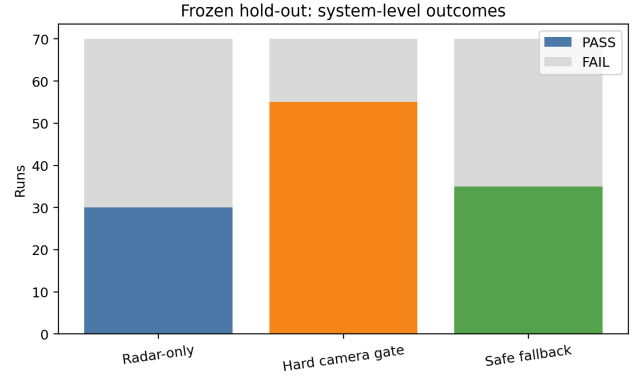
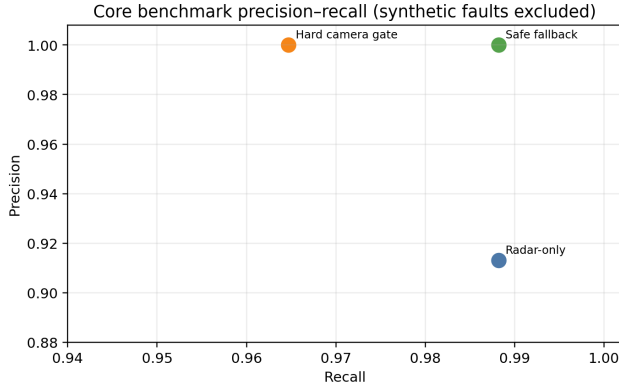


Fig. 2. Left: core precision–recall with synthetic faults excluded. Right: system-level PASS/FAIL on the frozen hold-out. Designed class ratios preclude road-prevalence interpretation.

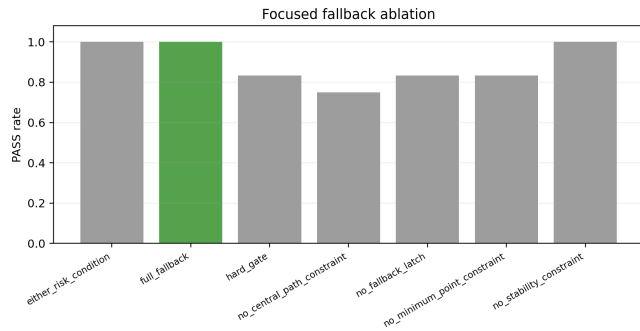


Fig. 3. Focused ablation. Full fallback passes 24/24; removing central-path, point-count, or latch constraints creates the indicated failures.

TABLE IV
Robustness and frozen hold-out outcomes.

Suite	Policy	Pass/N	TP	FP	TN	FN	Coll.
Perturb.	Radar	57/60	60	0	0	0	3
	Hard	45/60	45	0	0	15	15
	Fallback	54/60	60	0	0	0	6
Camera off	Hard	12/30	0	0	12	18	18
	Fallback	24/30	18	0	12	0	6
Hold-out	Radar	30/70	30	25	10	5	15
	Hard	55/70	25	0	35	10	15
	Fallback	35/70	30	20	15	5	15

C. Frozen hold-out and computational evidence

Hold-out is the decisive non-dominance result. All policies collide in 15 runs covering three new central props. The shopping cart produces at most two path candidates and no confirmed target; bench and warning props trigger too late. Fallback activates on four convincing central/0.50-m-offset ghosts, producing 20 false brakes, and blocks only the 0.75-m-offset ghost. Thus its hold-out precision/recall are 0.600/0.857 versus hard gate’s 1.000/0.714. In paired system status, hard gate alone passes 20 runs, fallback alone passes none, both pass 35, and both fail 15. The fallback recovers some brake labels but does not solve unknown-obstacle detection.

The campaign records 474 CUDA sessions, 74,928 inferences, and zero inference errors. Session-median p50/p95 are 9.95/11.53 ms and weighted mean is 11.56 ms. Each isolated process has one cold start above 150 ms; the maximum is 276.10 ms. These wall-clock measurements establish steady-state throughput on the tested GPU while disclosing startup cost. Because synchronous simulation waits for processing, they do not prove an automotive ECU deadline.

D. Failure-mechanism decomposition

The hold-out physical failures are not one homogeneous camera-gate effect. The shopping cart yields at most two in-path candidates and no confirmed cluster, so no policy brakes. The bench produces confirmed radar tracks; radar-only first brakes at a 12.62-m bumper gap, fallback waits for emergency evidence until 4.30 m, hard gate never brakes, and all collide. The traffic-warning prop causes a brake at 21.87 m even under hard gating, indicating apparent camera confirmation, yet all policies still record collision. These cases separate missing track, late emergency qualification, and collision/dynamics behavior.

The ghost results isolate a different mechanism. A six-point central injection reaches one confirmed cluster and triggers radar-only/fallback at 0.15 s, whereas hard gate blocks it. Eight- and ten-point central ghosts behave similarly. At a 0.75-m injected offset, radar-only still brakes but fallback rejects the track by its 0.65-m path criterion. The expected precision loss is therefore a direct consequence of fallback evidence, not YOLO timing.

The system-limit collisions are retained separately from perception trade-offs. All policies brake in the 15 repeated failures associated with CCRb 95/110 km/h at 20 m and the shortest high-speed cut-in; available distance is insufficient under the configured dynamics. The regression suite adds five no-brake late cut-outs at a lateral-corridor timing boundary. Conflating these with prop/ghost outcomes would obscure which component can plausibly be improved.

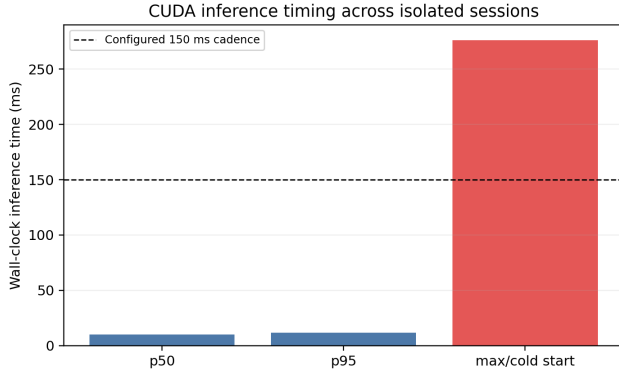


Fig. 4. CUDA processing: steady-state p50/p95 versus one disclosed cold-start outlier in each isolated detector process.

E. Traceability and statistical interpretation

Every child run records the Git commit, seed, model and configuration SHA-256, CARLA client/server version, fixed step, configured and active providers, and per-tick gate reasons. The complete archive includes raw CSV, JSON summaries, generated configuration snapshots, campaign/session manifests, and a SHA-256 sidecar. Re-running the analysis script regenerates the confusion, paired, scenario-consistency, ablation, sensitivity, and timing tables. This traceability is necessary because a scalar score cannot reveal whether a true-positive brake was camera-confirmed, fallback-triggered, or too late.

Run-level Wilson intervals quantify finite counts but do not make repeated runs independent. Scenario-level outcomes are therefore retained. The full66 core sweep has 63 all-PASS and three all-FAIL named scenarios for each policy; the mixed regression boundary is replaced by five consistent failures under fresh server isolation. In perturbation, each failed named condition fails all three repeats. In hold-out, every named condition is consistently all-PASS or all-FAIL. This consistency strengthens attribution to configured boundaries while reducing the effective independent sample size relative to the run count.

System-level paired comparisons also differ from brake confusion. On core, fallback versus hard gate has 525 both-PASS, ten fallback-only PASS, and 20 both-FAIL runs. On hold-out, it has 35 both-PASS, zero fallback-only PASS, 20 hard-only PASS, and 15 both-FAIL. Reporting both matrices prevents the additional fallback true-positive brakes in already-colliding prop cases from being misrepresented as safety improvements.

VI. Discussion and Limitations

A. Design implications

The three policies correspond to different error costs rather than a strict ranking. Radar-only is appropriate as a diagnostic upper bound on radar hazard recall, but its responses to every designed edge prop and synthetic fault make it unsuitable when unnecessary braking is costly. Hard gating is the most

conservative about unverified radar objects and has the highest hold-out system PASS count, but its car-only semantics make non-vehicle recall structurally limited. Emergency fallback deliberately moves toward radar recall and should be selected only if the operational design domain and downstream controller can tolerate its high-support-ghost failure mode.

The fallback conditions also suggest where rule tuning reaches diminishing returns. Point count, confidence, persistence, path offset, TTC, and stopping margin are all attributes of the estimated track; none is independent evidence that an object exists. Increasing their thresholds rejects weak faults but also consumes stopping distance. A qualitatively new evidence source is needed to break this symmetry: calibrated free-space estimates, broader obstacle semantics, radar cross-section or temporal micro-Doppler cues, multi-sensor existence probability, or map-independent occupancy. Until then, tuning shifts the operating point rather than eliminating the trade-off.

Evaluation should consequently report more than brake confusion. A policy may produce a true-positive brake yet collide because target formation or permission arrives too late. Conversely, a false brake has severity dimensions—onset speed, duration, deceleration, and traffic context—that a binary FP count does not capture. This study reports collision, first brake, minimum gap, and gate reason in raw logs, but its headline tables remain binary for comparability. Future campaigns should pre-register impact speed, residual stopping margin, brake duration, and jerk-aware cost, then aggregate by named condition before run-level totals.

B. Validity boundaries

The evidence supports a narrower conclusion than “fusion is better.” Camera gating is effective false-positive suppression when the detector label space covers the intended hazard. Radar fallback can recover selected out-of-label objects, but any rule that accepts a stable, central, high-support radar track is also vulnerable to a stable, central, high-support ghost. Raising quality or risk thresholds then delays or suppresses genuine-object braking, as sensitivity and camera-off results demonstrate.

External validity remains limited by one map and ego vehicle, favorable imaging, a one-class detector, and point-level CARLA radar. Synthetic ghosts are explicit fault models rather than measured multipath distributions. The hold-out adds appearance and prop diversity but is small and deliberately balanced. No hardware-in-the-loop, actuator/communication delay, tire-friction sweep, real sensor data, or vehicle test is included. The CCR families are inspired by AEB testing but do not reproduce Euro NCAP apparatus or certify compliance [10]. Future work should prioritize uncertainty-aware tracking, free-space or multi-class visual evidence, time-varying sensor degradation, recorded radar/camera data, and impact-speed/stopping-margin objectives before expanding nominal repeat counts.

VII. Conclusion

A conservative radar emergency fallback combines hard-gate precision and radar-only recall on the constructed core benchmark, but frozen hold-out results show that this combination is not general. Strong radar ghosts trade precision for recall, while some unseen physical obstacles are not tracked early enough by any policy. The main engineering outcome is therefore a reproducible map of trade-offs and failures: camera gating suppresses false brakes; emergency fallback conditionally restores obstacle recall; neither is a general safety solution. All claims are confined to synchronous CARLA simulation and do not constitute NCAP, functional-safety, real-time deployment, or real-vehicle validation.

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